

CET138 CSS: Cascading Style Sheets

Introduction

CSS (**Cascading Style Sheets**) is a style sheet language used to describe the visual presentation of HTML elements. While HTML provides structure and content, CSS controls how that content looks: colors, fonts, spacing, layout, animations, and responsive behavior.

CSS works through a system of selectors, properties, and values:

- Selectors: Target HTML elements (e.g., `h1`, `.classname`, `#idname`)
- Properties: What aspect to style (e.g., `color`, `font-size`, `background`)
- Values: What the property should be (e.g., `blue`, `16px`, `#2563eb`)
- CSS Variables: Custom properties that can be reused throughout stylesheets
- Responsive Design: Media queries that adapt layouts to different screen sizes
- Flexbox and Grid: Modern layout systems for complex page structures

Code Snippet CSS Variables & Navbar

```
/* CSS VARIABLES */
:root {
  --bg:      #ffffff;
  --bg2:     #f8f9fb;
  --accent:  #2563eb;
  --accent-dim: #dbeafe;
  --text:    #111827;
  --radius:  14px;
  --shadow:   0 4px 12px rgba(0,0,0,0.08);
}

/* NAVBAR */
#mainNav {
  background: #ffffff;
  backdrop-filter: blur(24px);
  border-bottom: 1px solid var(--border);
  transition: all 0.28s cubic-bezier(0.4,0,0.2,1);
}

.navbar-nav .nav-link:hover,
.navbar-nav .nav-link.active {
  color: var(--accent) !important;
  background: var(--accent-dim);
}
```

Explanation of Code

- `:root`: A pseudo-class that selects the document root. Variables declared here are globally available.
- CSS Variables (`--accent: #2563eb`): Custom properties referenced using `var(--accent)`. Enables easy theme changes.

- `#mainNav`: ID selector targeting the element with `id="mainNav"`.
- `backdrop-filter: blur(24px)`: Modern CSS effect creating a frosted glass background.
- `transition: all 0.28s ...`: Smoothly animates all property changes over 0.28 seconds.
- `.navbar-nav .nav-link`: Descendant combinator targeting `nav-link` inside `.navbar-nav`.
- `:hover` pseudo-class: Applies styles when the user hovers over an element.
- `!important`: Forces a declaration to apply regardless of specificity (use sparingly).

Demonstration of Understanding

- CSS Variables: Custom properties enable maintainable, theme-friendly stylesheets.
- Modern Effects: Backdrop filters and smooth transitions for polished UX.
- Specificity and Cascading: Understanding how selectors interact and override each other.
- Hover States: Interactive visual feedback when users interact with elements.
- Consistency: Predefined variables ensure consistent spacing, colors, and effects.

Conclusion

CSS is the layer between structure (HTML) and interactivity (JavaScript). Mastering CSS means understanding selectors, the box model, layout systems (Flexbox, Grid), and modern techniques like CSS variables and animations. Well-written CSS makes applications visually appealing while keeping code organized and maintainable.